

Technology Management - NUIST

Engineering Technology

May, 2026

Technology Management - NUIST (A35612)

Short Title: Technology Management - NUIST
Faculty Science and Computing
Department Engineering Technology
Cluster Engineering Practise
Author CLAIRE.FITZPATRICK
Credits 5

Level: Postgraduate

Description of Module / Aims

The aim of this module is to prepare individuals with a strategic mind-set, relevant to the biomedical sector of Ireland, fine tuned communication skills, and the ability to drive enhanced performance through the use of technology. Complex case studies involving organisational theory, conflict resolution and effective strategic management are core to this module.

Programmes

ENGR-0040	Certificate in Biomedical Engineering (WD_CBIOM_MA)	1 / 2 / M
ENGR-0040	MEng in Electronic Engineering (WD_EELEN_R)	1 / 2 / E
	MEng in Electronic Information Systems (WD_EELEN_RI)	3 / 6 / M
ENGR-0040	MSc in Innovative Technology Engineering (WD_CINNT_R)	1 / 2 / M
ENGR-0040	PG Dip in Science in Innovative Technology Engineering (WD_CINNT_G)	1 / 2 / M

Indicative Content

- Engineering and Technology Management: The management process: Empowerment and effective leadership
- Engineering and Technology Management: The regional & Global Context: Sources of competitive advantage: Insights into professional imperatives
- Ethics in Engineering / Technology
- Intellectual Property: Creating, Application and Protection: Learning Organisations
- Manufacturing and Production Management: Strategies and Tools
- Manufacturing & Production Management: Strategic importance of maintenance and reliability
- Career Development in a technology company. Managing your boss, peers and subordinates

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Using Kotter's Eight Step Model, synthesise change management tools specific to the optimisation of R&D processes in high value organisations.
2. Apply the knowledge and comprehension gained in the analysis of complex Enterprise Technology Management case studies to provide best in-class frameworks for SME and MNC organisations.
3. Evaluate complete and/or partial management strategies in respect of the research, design and manufacture of modern technological products; both with and without the aid of industry standard software support tools.
4. Synthesise complete and/or partial management strategies in respect of the research, design and manufacture of modern technological products; both with and without the aid of industry standard software support tools.

Learning and Teaching Methods

- Lectures
- Case Studies
- Assignments / Assessments
- Problem Based Mini Project

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Group Project	40%	1,2,3,4
Essay	50%	1,2,3,4
Participation	10%	1,2

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical		
Independent Learning	111	

Essential Material

- Griffin, R.W. *_Management_*. USA: Houghton Mifflin, 2008.

Supplementary Material

- Betz, F. *_Strategic Technology Management_*. New York: McGraw Hill, 1997.
- Tavani, T.H. *_Ethics & Technology_*. 2nd ed.. USA: Wiley, 2007.